PROBLEM-BASED LEARNING

Design and Construction of a Variable Regulated DC Power Supply

SUBMITTED BY:

- *SALMAN RAZA(CS-043)*
- *AHSAN ATIF RAJPUT(CS-O44)*
- *NIKEETA KUMARI(CS-046)*
- *ZAIN (CS-048)*
- *SYEDA MAHAM(CS-054)*
- *BUSHRA SEHAR(CS-068)*

SUBMITTED TO:
MISS ANILA ABBAS

Table of
CONTENTS

Table of
CONTENTS

Abstract

This project focuses on the design and construction of a regulated DC power supply with an adjustable output voltage ranging from 0V to 15V and a maximum current of 3A. The power supply converts a standard 220-230V AC input to DC using a step-down transformer, bridge rectifier, and a smoothing capacitor. The LM317 voltage regulator provides stable voltage output, which can be fine-tuned using a potentiometer. The circuit also incorporates capacitors to minimize ripple and transient currents. This power supply is ideal for low-voltage applications in labs or electronics projects.

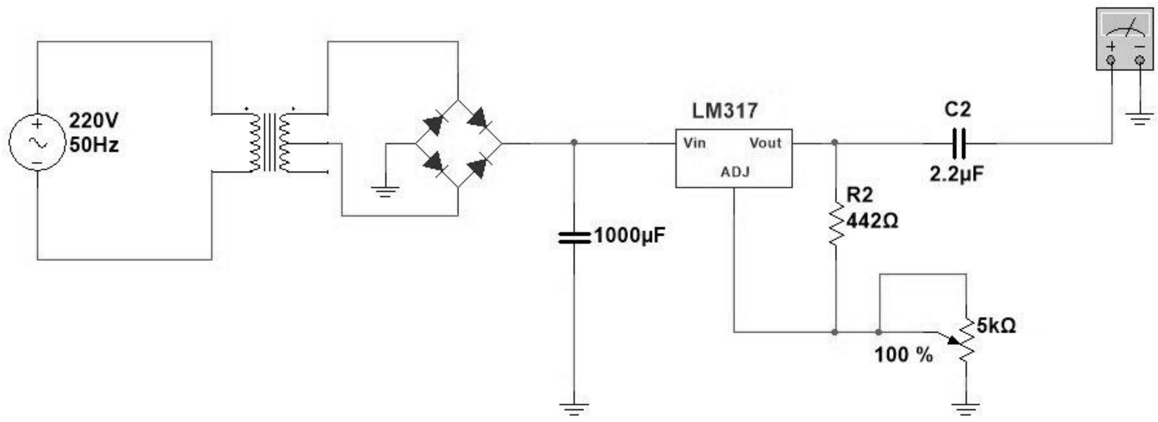
OBJECT:

Make a regulated DC power supply with variable voltage from 0v to 15v and 3A maximum current. The input is 220v 50Hz.

EQUIPMENT:

1. Step Down Transformer (12+12)v
2. Bridge Rectifier IC
3. Voltage Regulator LM317
4. Smoothing/Filter Capacitor 1000 Micro Farads, 50volts
5. 442 Ohms and 1.8 K ohms Resistors
6. Potentiometer 5K ohms
7. Output Capacitor 220 micro Farad,50v
8. Wires for connection
9. Bread board
10. LED 3v
11. Volt Meter display
12. Power Cord
13. Two pin connectors

CIRCUIT DIAGRAM:



THEORY OF THE COMPONENTS USED:

1. Step Down Transformer (12+12)V:
    a. **Definition**: A transformer that reduces the input AC voltage from a higher level (e.g., 230V) to a lower level (12V + 12V).
    b. **Purpose**: Converts high-voltage AC from the mains to a safer, lower voltage AC (12V) for further processing in the power supply circuit.
2. **Bridge Rectifier IC :**
    a. **Definition**: A diode-based rectifier IC that converts alternating current (AC) to direct current (DC).
    b. **Purpose**: Converts the AC from the transformer into unregulated DC, necessary for powering DC components.
3. **Voltage Regulator IC (LM 317):**
    a. **Definition**: An adjustable voltage regulator that outputs a stable DC voltage despite fluctuations in input voltage or load conditions.
    b. **Purpose**: Maintains a constant and adjustable output voltage, providing a smooth and regulated DC supply to the load.

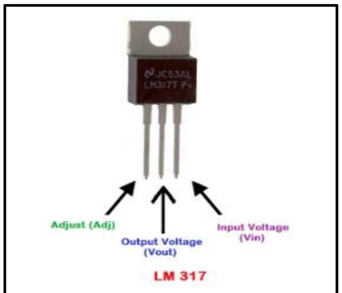
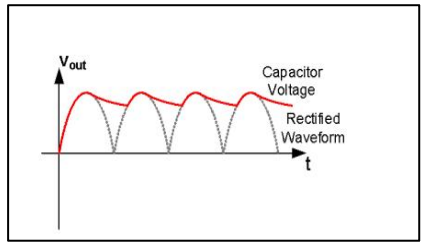


Figure 1integrated chip 317

4. **Smoothing/Filter Capacitor (1000 µF, 50V):**
    a. **Definition**: A large electrolytic capacitor used to filter out ripples in the DC signal.
    b. **Purpose**: Smooths the rectified DC by reducing voltage fluctuations (ripples) from the bridge rectifier, resulting in a more stable DC output.



5. **Resistors (442Ω and 1.8 KΩ):**
    a. **Definition**: Components that limit current flow and divide voltage in a circuit.
    b. **Purpose**: Used for setting the output voltage and current regulation in the circuit when used with the LM317 voltage regulator.
6. **Potentiometer (5KΩ):**
    a. **Definition**: A variable resistor that adjusts resistance in a circuit.
    b. **Purpose**: Allows the adjustment of output voltage from the voltage regulator, enabling fine-tuning of the output.

Figure 3 potentiometer

7. **Output Capacitor (220 µF, 50V):**
    a. **Definition**: A capacitor connected near the output to further filter the voltage.
    b. **Purpose**: Ensures smooth DC output by filtering any residual noise or ripples after regulation.

Figure 5 capacitor

8. **Wires for Connection:**
    a. **Definition**: Conductive materials used to connect components.
    b. **Purpose**: Provides the physical electrical connections between components in the

Figure 6 Connecting wires

circuit.

9. **Breadboard:**
    a. **Definition**: A reusable platform for building and testing circuits without soldering.
    b. **Purpose**: Holds and connects the components temporarily for prototyping and testing the power supply design.

10. LED (3V):

- **Definition**: A light-emitting diode that produces light when current flows through it.
- **Purpose**: Acts as an indicator to show whether the circuit is powered and working correctly.

11. **Voltmeter Display:**

- **Definition**: A device that measures and displays the voltage in a circuit.
- **Purpose**: Allows real-time monitoring of the output voltage from the power supply.

12.Power Cord:

- **Definition**: A cable that connects the power supply to the mains electricity.
- **Purpose**: Supplies AC power to the step-down transformer from the wall socket.

13.Two Pin Connectors:

- **Definition**: Connectors used for easy plugging and unplugging of wires.
- **Purpose**: Provides a convenient way to connect or disconnect the power supply from the circuit or load.

PROCEDURE:

1. Connect the 220V transformer which generates the AC signal with the rectifier.
2. The rectifier converts the AC signal into pulsating DC
3. To convert the pulsating DC into pure DC or constant DC attach the rectifier to the filter.
4. The filter is then connected to the voltage regulator which gives the adjustable desired voltage and removes leftover ripples.
5. The potentiometer which is connected to the regulator to provide a variable voltage.
6. The voltage regulator is then connected to a load which is a resistor of 1.8kohms

WORKING FORMULA:

V(peak AC)=V(rms) x 1.414
V(peak DC)=V(peak AC)-1.4

CALCULATIONS:

V(peak AC)= 12x 1.414
=16.9volts

V(peak DC)= 16.9-1.4
=15.5volts

OBSERVATIONS:

a) AC input and Frequency is 220-230 volts and 50 Hz.

b) Peak Voltage as output from transformer is 16.9 volts.

C)Peak voltage as output from rectifier is 15.5 volts

MAXIMUM VOLTAGE OUTPUT	MINIMUM VOLTAGE OUTPUT
15	1.21

RESULT:

The Max Output voltage = Vo = 15.3 volts.
The Max Output current using load resistor = Io = 0.72 Ampere.